



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	NUCLEAR PHYSICS				
Type of Course	Major Elective				
Semester	VIII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	PHY6CJ303: NUCLEAR AND PARTICLE PHYSICS				
Course Summary	This course explores advanced nuclear and particle physics.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Interpret the properties of nucleus, binding energy, angular momentum, two-nucleon scattering, spin dependence, tensor force, partial wave concept and theory of deuteron structure	An	C	Instructor-created exams / Quiz
CO2	Elucidate the theory of various types of nuclear decay, selection rules of transition, concept of parity and multipole moments.	U	C	Instructor-created exams / Quiz
CO3	Comparison of various nuclear models.	An	P	Instructor-created exams / Home Assignments

CO4	Comparison of nuclear processes like fission and fusion and the concept of nuclear reactor.	An	P	Instructor-created exams / Home Assignments
CO5	Demonstrate the working of one or two nuclear radiation detectors of different types	Ap	P	Seminar Presentation / Group Tutorial Work
CO6	Compare basic interactions and classify the elementary particles. Interactions are linked with the concept of symmetry and conservation laws. Understand Sakata model, Gellmann- Okubo mass formula, Quark mode and their significance.	An	P	Seminar Presentation / Group Tutorial Work / Group Project
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F), Conceptual Knowledge (C), Procedural Knowledge (P), Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Mar ks
I	NUCLEAR FORCES		13	18
	1	Basic nuclear properties: nuclear size, shape, mass, binding energy, angular momentum and parity	3	
	2	Simple theory of deuteron	3	
	3	Low energy n-p scattering	2	
	4	Properties of Nuclear force	2	
	5	Scattering cross section, phase shift	1	
	6	Singlet and triplet potentials	1	
	7	p-p and n-n scattering	1	
	Sections 8.1 – 8.6 of chapter 8 of Book 1			
II	NUCLEAR MODELS, FISSION AND FUSION		12	18

	8	Shell model, Spin-orbit potential, Magnetic dipole and electric quadrupole moment, spin and parity	3	
	9	Collective structure, Nuclear Rotation and vibration	2	
	10	Liquid drop model and Semi-empirical mass formula	2	
	11	Energy and Characteristics of fission	1	
	12	Fission reactors	2	
	13	Basics and Characteristics of fusion	1	
	14	Solar fusion	1	
	Sections 5.1-5.2, 13.1-13.3,13.6, 14.1-14.3 of Book 1			
III	NUCLEAR DECAY, FISSION AND FUSION		13	18
	15	Theory of alpha decay	2	
	16	Energy Release in beta decay	1	
	17	Fermi theory of beta decay	2	
	18	Angular momentum and parity selection rules	2	
	19	Neutrino Physics	2	
	20	Energetics of Gamma decay	1	
	21	Angular momentum and parity selection rules	2	
	22	Internal Conversion	1	
	Sections 8.1-8.4, 9.1,9.2,9.4,9.6, 10.1,10.4,10.6 of Book 1			
IV	NUCLEAR DETECTORS		10	16
	23	Gas detectors	3	
	24	Scintillation Counter	2	
	25	Semiconductor detectors	3	
	26	Single channel analyser	1	
	27	Multichannel analyser	1	
	Relevant sections of Book 2			
V	OPEN-ENDED MODULE: PARTICLE PHYSICS		12	

Books and References:

1. Kenneth S Krane : Introductory Nuclear Physics (Wiley)

2. S S Kapoor and V S Ramamurthy : Nuclear Radiation Detectors (Wiley)

3. G F Knoll : Radiation Detection and Measurement (Fourth Edition, Wiley, 2011)

4. B.L.Cohen : Concepts of Nuclear Physics (Tata McGraw Hill)

5. S.B.Patel : An Introduction to Nuclear Physics (New Age International Publishers)
6. D C Thayal : Nuclear Physics (Himalaya Publishing House)

Mapping of COs with PSOs and POs:

	PSO 1	PSO 2	PSO 3	PSO 4	PSO 5	PSO 6	PO 1	PO 2	PO 3	PO 4	PO 5	PO6	PO7
CO 1	3	-	2	-	2	-	3	-	1	-	2	-	-
CO 2	2	3	3	-	2	2	2	-	2	-	2	-	-
CO 3	2	2	2	-	2	2	1	-	1	-	1	-	-
CO 4	2	2	2	-	2	2	1	-	1	-	1	-	-
CO 5	-	2	3	1	2	-	-	-	1	-	1	-	1
CO 6	-	3	-	1	2	2	-	-	1	-	1	-	1

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Assignments
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Practical / Computational Skill Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓			✓
CO 4		✓		✓
CO 5		✓		✓
CO 6			✓	

MINOR COURSES